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TECHNICAL REPORT 05

EnergieBee core Predbat slot compatibility

How the core's time grid lines up with Predbat

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Contents

Verdict 3

Example 1: the grid 4

Example 2: an off-peak window 4

Example 3: a price over a slot 5

Where Predbat's tables would attach 5

What would need building 5

Content is reproduced from the source document `docs/predbat-slot-compatibility.md`, which remains the editable original.

REPORT OVERVIEW

How the core's time model lines up with Predbat's step, slot, window and horizon. Source for the Predbat side: the "Predbat and the Slot Model" document. Predbat itself is not built in the core: only the time grid, prices and facts it needs are.

Verdict

Predbat idea	In the core	Status
5-minute **step**	5-minute inverter readings; the build integrates them into slots	Compatible
Slot (default 30 min, configurable)	`site.slot_s` = 900, 1800 or 3600; each ledger row stores `slot_start`, `dur_s`, `tod_min`	Compatible
Slots keyed by UTC epoch start	Same: `slot_start` is epoch seconds, aligned to the slot length	Compatible
46 / 50 slots on clock-change days	Tested for London, Paris and Amsterdam at 15, 30 and 60 min	Compatible
Local tariff windows resolved to UTC per date	`localToUtc` and the window expander, clock-change safe	Compatible
Price array per slot	`rate_interval` rows with `priceOver` (time-weighted when a slot spans prices)	Compatible
Dispatch slots (Intelligent Octopus)	Counted against the daily allowance; slots starting at :15 or :45 are covered	Compatible
48-hour **horizon**	Forecast for 48 hours exists; no plan over it	Partial
Windows (merged slots)	Tariff windows only; plan windows not built	Missing
Five **actions** per slot	No plan table	Missing
Export start offset (0 to 25 min)	Not built. Rule for other slot lengths: offset is a multiple of 5 min and less than the slot	Missing
Write only on change, read back	No device control	Missing
Stale inputs give a demand-only plan	Data states exist (final, provisional, missing); no planner gate	Missing
Two inverters planned as one battery	Not handled	Missing

Example 1: the grid

A home in London at 14:37:20 local time on 23 October 2026 (13:37:20 UTC, clocks still on summer time).

Predbat quantity	At 30-min slots
`now`	2026-10-23T13:37:20Z
`plan_start` (rounded down to the slot)	2026-10-23T13:30:00Z
`minute_offset` into slot 0	7 minutes
First simulation step	step 1 (the step that starts at :35)

Slots per day in London, from the core's own time code:

Slot length	Normal day	25 October 2026 (clocks go back)	28 March 2027 (clocks go forward)
15 min	96	100	92
30 min	48	50	46
60 min	24	25	23

Steps per slot are `slot_s / 300`: 3 at 15 minutes, 6 at 30, 12 at 60. Predbat's formulas stay the same, with that one number changed.

Example 2: an off-peak window

Predbat's "off-peak 23:30 to 05:30" starting the evening of 23 October 2026, London.

	Value
Start, local	23:30
End, local	05:30 on 24 October
In UTC	2026-10-23T22:30:00Z to 2026-10-24T04:30:00Z
Slots at 30 min	12
Slots at 15 min	24

The core resolves the window per date with the home's time zone, as the Predbat document requires. On the night of 24 to 25 October the same window is 7 real hours long, not 6, because the clocks go back.

Example 3: a price over a slot

A 30-minute slot with two 15-minute prices (illustrative numbers, not real tariffs):

Quarter	Price	Import
10:00 to 10:15	10 c/kWh	0.1 kWh
10:15 to 10:30	30 c/kWh	0.5 kWh

- At 15-minute slots the cost is exact: 0.1×10 + 0.5×30 = 16 c.
- At a 30-minute slot the core uses the time-weighted price, 20 c/kWh, so 0.6 kWh costs 12 c.
- The 4 c difference is the price of a longer slot than the price interval. The default slot is therefore the finest price interval of the tariff.

Where Predbat's tables would attach

Predbat input	In the core today
Rates per slot	`rate_interval` joined to `tariff_agreement`
PV forecast (central and pessimistic)	`forecast_fact` issues (central only today)
Load history	`site_hh` and 5-minute `reading` rows
Live telemetry	`reading`
Plan, plan slots	Not built (`plan_run`, `plan_slot` are designed in the roadmap)

A plan slot would join to what actually happened on the same key: `plan_slot.slot_start = site_hh.slot_start`.

What would need building

1. `plan_run` and a device-generic `plan_slot` (device, slot, action, parameters), stored only when the plan changes.
2. The simulator, optimiser and window merge as a pure function of the facts known at a given moment.
3. The command path with expiry and read-back.

See `docs/capabilities-and-roadmap.md` for the full list.